GRADE 12 PHYSICS STUDY GUIDE

Federal Democratic Republic of Ethiopia • Ministry of Education Curriculum

Unit 1: Application of Physics in Other Fields

Physics provides the baseline physical laws, analytical framework, and instrumentation that power related scientific disciplines and engineering branches.

Physics in Natural Sciences

• **Chemistry:** Explains atomic structures, chemical bonding, electron energy levels, and spectroscopy.
• **Biology:** Governs biomechanics, blood flow dynamics, nerve signal conduction, and biological optics.
• **Astronomy:** Astrophysics utilizes celestial mechanics, stellar radiation, and spectroscopy.
• **Geology:** Geophysics analyzes seismic wave transmission, gravity anomalies, and Earth dynamics.

Engineering & Medicine

• **Civil/Mechanical:** Rely on Newtonian mechanics, statics, materials science, and thermodynamics.
• **Electrical:** Built upon Maxwell's electromagnetic equations and solid-state physics.
• **Medical Imaging:** Utilizes diagnostic tools like X-rays, MRI (magnetic resonance), CT scans, Ultrasound, and Radiotherapy.

Unit 2: Two-Dimensional Motion and Gravitation

Two-dimensional motion models objects moving simultaneously along horizontal and vertical axes under constant forces like gravity.

Projectile Motion Principles

Assuming air resistance is negligible, horizontal motion (x) and vertical motion (y) are completely independent. Horizontal velocity ($v_x = v_0 \cos \theta$) remains constant, while vertical velocity ($v_y = v_0 \sin \theta - gt$) changes under constant gravitational acceleration (g).

Key Projectile Equations:

• Horizontal Position: $x = (v_0 \cos \theta)\, t$
• Vertical Position: $y = (v_0 \sin \theta)\, t - \frac{1}{2} g\, t^2$
• Maximum Height: $H = (v_0^2 \sin^2 \theta) / (2g)$
• Horizontal Range: $R = (v_0^2 \sin 2\theta) / g$

Rotational Kinematics Analogy

Rotational motion around a fixed axis directly parallels linear kinematics through angular variables measured in radians:

• Displacement: $s = r\, \theta$
• Velocity: $v = r\, \omega$

Unit 2 (Cont.): Dynamics & Gravitation

Rotational Dynamics & Kepler's Laws

Torque ($\tau = r\ F\ \sin\theta$) produces angular acceleration according to $\tau = I\ \alpha$, where I is the moment of inertia. Newton's Law of Gravitation ($F_g = G\ m_1\ m_2 / r^2$) provides the theoretical foundation for Kepler's laws of planetary motion.

Law	Statement & Physical Implications
Kepler's 1st Law	All planets move in elliptical orbits with the Sun at one focus.
Kepler's 2nd Law	A line connecting a planet and the Sun sweeps out equal areas in equal time intervals (Conservation of Angular Momentum).
Kepler's 3rd Law	The square of the orbital period is proportional to the cube of the semi-major axis: $T^2 / R^3 = 4\pi^2 / (G\ M)$.

Unit 3: Fluid Mechanics

Fluid mechanics explores the physical behavior of liquids and gases in static conditions and steady flow motion.

Fluid Statics

• **Pressure:** Defined as force per unit area ($P = F / A$). Static fluid pressure at depth h is given by $P = P_0 + \rho\ g\ h$.
• **Pascal's Principle:** Pressure applied to an enclosed fluid is transmitted undiminished throughout the fluid ($F_1 / A_1 = F_2 / A_2$).

Archimedes' Principle

Any body completely or partially submerged in a fluid experiences an upward buoyant force equal to the weight of the fluid displaced:
$F_b = \rho_{fluid}\ V_{displaced}\ g$

Fluid Dynamics & Flow Continuity

For an ideal, incompressible, and non-viscous fluid undergoing steady flow, fluid speed varies inversely with cross-sectional area as defined by the **Equation of Continuity**:

$$A_1\ v_1 = A_2\ v_2 = \text{Constant Flow Rate } (Q)$$

Unit 4: Electromagnetism

Electromagnetism links electrical charges and magnetic fields through electromagnetic induction, described mathematically by Faraday's and Lenz's laws.

Faraday's & Lenz's Laws

• **Magnetic Flux:** $\Phi_B = B\,A\cos\theta$

• **Faraday's Law:** Induced emf is directly proportional to the rate of change of magnetic flux:
$\varepsilon = -N\,(\Delta\Phi_B / \Delta t)$

• **Lenz's Law:** The direction of induced current opposes the change in magnetic flux that produced it.

Transformers

Transformers alter alternating voltage levels using mutual induction across isolated coils:
$V_p / V_s = N_p / N_s = I_s / I_p$
Used extensively in power grids, electric motors, relays, and generators.

Unit 5: Basics of Electronics

Modern electronic circuits control current flow using solid-state semiconductor devices and binary logic operations.

Semiconductors & P-N Junctions

Pure silicon/germanium is doped with impurities to create **N-type** (excess free electrons) or **P-type** (excess electron holes) materials. A **P-N junction diode** permits current flow only in forward bias, serving as an AC-to-DC rectifier. Bipolar Junction Transistors (NPN/PNP) function as electronic switches and current amplifiers.

Fundamental Logic Gates

Gate	Symbol / Formula	Logic Operation Output
AND	$Y = A \cdot B$	Output is 1 ONLY if all inputs (A and B) are 1.
OR	$Y = A + B$	Output is 1 if at least one input (A or B) is 1.
NOT	$Y = \overline{A}$	Inverter: Output is the opposite of the single input.